XRPL Validator — Daily Engineering Report · 2026-10-30

Period: 2026-10-30 00:00 to 24:00 US Eastern (2026-10-30 04:00 to 2026-10-31 04:00 UTC); ledgers #107,993,768 to #108,015,363. Every count from the ledger and the validator's logs stops at #108,015,363, closed 03:59:52 UTC. Generated 2026-10-31 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded — 1 engine receipt open.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,592 of the 21,596 ledgers the network closed (99.981%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 4 outside maintenance (1 incident, 1 isolated miss).
- **Native engine:** 2 engine receipts: #108,008,726 — fixed: commit cccdec7 (2026-10-30); #108,011,720 — triaged: an isolated replay reproduced the difference.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Independence program:** 2 of 30 qualifying days (snapshot at 04:20:46 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,596
Validations issued	21,592
Coverage (validations / ledgers closed)	99.981%
Validations matching the network's validated ledger hash	21,592 of 21,592
Unsigned during planned restarts	0
Unsigned outside maintenance	4
Coverage outside planned maintenance (objective 99.9%)	99.981%
Restarts	0
Basis: our own ledger hash	21,592
External agreement, data.xrpl.org (UTC day 2026-10-30)	0.99990 (2 missed of 20,630)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,993,768–#108,015,363), by step:

Step	Ledgers with a receipt
Step 1 · native engine	2
Step 2 · records	2
Step 3 · order	2
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	2
Writer	2

Live counters, snapshot at 04:20:46 UTC (since the last restart, 2026-10-28 22:52 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	47,538 ledgers, 9,877,475 transactions	47,537 fully matched; results 2, bytes 1, keys 0/0
2 · transaction records	47,959 ledgers, 10,061,870 records	10,061,867 exact; differ 3, tree 3
3 · transaction order	47,998 ledgers (mode: on)	47,996 exact; order 0, results 2, bytes 0, refused 0
4 · ledger data from peers	48,047 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	48,043 ledgers judged	48,043 agreed; differ 0, short of quorum 0
5 · our own ledger hash	48,046 ledgers	47,954 own; differ 3, partial 89
Writer	every ledger written	mismatch 1, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	20:40 UTC	#108,008,726	Step 1 · native engine, Step 2 · records, Step 3 · order, Step 5 · own hash, Writer	AMMCreate tx 34C1C80B45F6: the transaction's result differed (ours=162 theirs=137)	fixed: commit cccdec7 (2026-10-30); awaiting merge and deployment; see INC-01
R-02	00:00 UTC	#108,011,720	Step 1 · native engine, Step 2 · records, Step 3 · order, Step 5 · own hash, Writer	AMMCreate tx 35B6EB4B0913: the transaction's result differed (ours=162 theirs=137)	triaged: an isolated replay reproduced the difference; no fix recorded for this ledger

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-30-01, INC-01 is INC-2026-10-30-01.

Incidents

INC-01 · 2 ledgers unsigned

Window	20:40:02 UTC to 20:40:10 UTC
Ledgers	#108,008,726–#108,008,727 (2)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-01
Cause	#108008726 34C1C80B45F6: a PLX/BEEP AMMCreate by an account whose BEEP line its issuer froze (lsfLowFreeze, the issuer being the low side). rippled's preclaim runs requireAuth on both amounts, then checkFrozen on both, then terNO_RIPPLE for an issuer without lsfDefaultRipple, all before the reserve and funds (AMMCreate.cpp:107-146). We had none of the three, so the frozen side counted as zero funds and we answered tecUNFUNDED_AMM where mainnet answered tecFROZEN. Fee-only either way, but the result code is in the metadata, so our transaction tree differed and that ledger went unsigned on the …
Action	commit cccdec7 — fix(amm): AMMCreate checks auth, freeze and DefaultRipple before funds; awaiting deployment

Isolated unsigned ledgers (shorter than 3 ledgers, cause not recorded): #108,011,720–#108,011,721 (2) at 00:00:10 UTC.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **2 of 30** (snapshot at 04:20:46 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-30 — tracked #107,990,191–#108,011,717	Value
Validations	21,525 (21,525 own, 0 borrowed)
Coverage of the tracked ledgers	99.99%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,596 (#107,993,768–#108,015,363)
Mean close interval	4.00 s
Longest close interval	10 s (before #108,015,227)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Receipt R-02 (#108,011,720): open, under investigation.
- Receipt R-01 (#108,008,726): fixed: commit cccdec7 (2026-10-30); awaiting merge and deployment; see INC-01.
- Unsigned #108,011,720–#108,011,721 (2) (00:00:10 UTC): cause not recorded.
- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.